STAT 525: Topics in Computational Statistics
Fall 2024

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Course Information

Course Overview

Computational data analysis is an essential part of modern statistics and machine learning. This course covers topics to develop scalable computational tools for modern statistics and data analysis. It is designed to help students develop a practical understanding of how and why the existing methods work so that students can learn the core ideas of these methods and use modern statistical methods effectively. This course emphasizes theory and algorithms for smooth convex optimization and non-smooth convex optimization, stochastic convex optimization, convex finite sum optimization, and their applications in statistics, including Lasso, group Lasso, total variation methods, and large-scale machine learning methods. This course doesn't focus on programming skills training, but rather on the computation complexity theories and ideas behind algorithms. This course will also provide opportunities for exploring frontier research in optimization and machine learning.

Textbooks

You can order the required textbook from the Illini Union Bookstore (IUB) (217-333-2050). Be certain you select the correct term from the Search by Course dropdown list. Used books are often available and can be provided at a cost savings to you. Note that the cost and postage for all books must be prepaid. You can charge your order using VISA, Discover, or MasterCard. If you are ordering books elsewhere (e.g., Amazon.com), please be sure to order the correct edition and year of publication.

- Optimization for Data Analysis, by Wright & Recht (Recommended)
- First-order and Stochastic Optimization Methods for Machine Learning, by Lan (Optional)
- Convex Optimization, by Boyd & Vandenberghe (Optional)
- First-Order Methods in Optimization, by Beck (Optional)

- Lectures on Convex Optimization, by Nesterov (Optional)
- Numerical Optimization, by Nocedal & Wright (Optional)
- High-Dimensional Data Analysis with Low-Dimensional Models: Principles, Computation, and Applications, by Wright & Ma (Optional)

Readings and Lectures

Each week will contain a list of lectures and assigned readings. In some cases, optional or supplemental readings may be listed for further study. Lectures cover major topics from the readings but do not necessarily include all important information from the readings.

Topic Outline

1. Statistical and Machine Learning Models
2. Convex Optimization Theory, including Convex Sets, Convex Functions, Subgradient, Lagrange Duality, KKT Conditions, and Conjugate Function.
3. Deterministic Smooth Convex Optimization, including Gradient Descent, Projected Gradient Descent, Conditional Gradient Descent, Accelerated Gradient Descent, Conjugate Gradient Method, Coordinate Descent, and ADMM.
4. Deterministic Non-smooth Convex Optimization, including Subgradient Descent, Mirror Descent, Proximal Descent Method, Smoothing, Dual Proximal Gradient Method, Primal-Dual Subgradient Method, and Mirror Prox Algorithm.
5. Stochastic Optimization, including Sample Average Approximation, Stochastic Approximation/Stochastic Gradient Descent, Stochastic Mirror Descent, Stochastic Variance Reduced Gradient, AdaGrad, RMSprop, and Adam.

Prerequisite

Stat 510, linear algebra, analysis, and some experience with programming. The R programming language is needed for completing the assignments and exams.

Grading

- Bi-weekly homework assignments (60%): The lowest score of 6 homework will be dropped.
- Final project (40%): The students need to give a presentation in the last two weeks of the semester.

Grade Disputes

If you feel your work was graded incorrectly or the score on Canvas is reflected incorrectly,

- you have one week from the date the score was posted on compass(an announcement will be posted on Canvas when grades are up)
- After one week, grading is final.
- You may not simply ask for a re-grade, but instead must justify in writing to the staff why you feel grading was done incorrectly by stating clearly the question(s) you would like regraded and providing an explanation of why you feel it is graded incorrectly.
- By disputing any grading, you agree to allow the course staff to review all problems for other errors missed during grading.

Grading Scale

The percentages in the table below will guarantee the corresponding letter grade.

A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-
TBD	93%	90%	87%	83%	80%	77%	73%	70%	67%	63%	60%

Any rounding or curving will not be decided till after finals week when all grades are in. Please do not ask me after each exam or before the end of the semster if there is any curve. I will have to refer you to this part of the syllabus.

Course Policies

Accommodations

To obtain disability-related academic adjustments and/or auxiliary aids, students should contact both the instructor and the Disability Resources and Educational Services (DRES) as soon as possible. You can contact DRES at 1207 S. Oak Street, Champaign, (217) 333-1970, or via email at disability@illinois.edu.

Participation

Active participation in the learning environment is vital to your success in this course. Depending on your course, you may be asked to engage in discussions and other interactive learning environments that invite your active participation and involvement with other students and your instructor.

Student Commitment

By registering for this course, you commit to self-motivated study, participation in course activities, and timely submission of all assignments. Furthermore, you commit to accessing the course website and checking email at least twice per week, as well as to devoting at least 8–12 hrs./week to preparing for each module and completing the required assignments and readings.

Deadlines

If you are unable to meet a particular deadline, it is your responsibility to make prior arrangements with the instructor for that given week. Otherwise, work submitted later than deadline will receive 80% of grades, and work submitted later than 2 days will not be considered for grading.

Homeworks

Homework will be turned in on Canvas. Homework assignments to be turned in on Canvas will require you to format your assignment and code in Rmarkdown, knit the file and submit on Canvas by following all naming and formatting directions. Please refer to the homework policy document for all detailed information about formating, naming files, submiting and a rough grading rubric for your assignments on Canvas. **WARNING:** Do not wait till the last day. The time on the computer is not necessarily the time on your watch. Manage your time adequately and begin problems early and come back to them often.

Please come talk to me if there are difficulties; problems/conflicts must be discussed IN ADVANCE, preferably in the first week of classes or as they arise and no later than noon on the day before the homework is due. Emergencies on the day the homework is due that you did not know of by noon on the day before the homework due date must be verifiable. If the emergency cannot be verified, the student is left to deal with the late homework policy. Such emergencies must be discussed in person, pereferably during office hours. Private emails for such matters will be ignored.

Responding to Emails (only for questions of personal nature)

Private emails to the TAs and Professor should sent for truly private matters. Private emails that ask questions about course materials, course administration or homework questions may be sent, but replies are not guaranteed. Due to the large size of this course, we follow a strict email policy as detailed in course polices.

In seeking help, please use office hours. For questions of personal nature, please see me or course staff during office hours. If an online meeting is absolutely not possible given the personal situation, please state so in your private email to me or the course staff as appropriate.

Academic Integrity

Academic dishonesty will not be tolerated. Examples of academic dishonesty include the following:

- Cheating
- Fabrication
- Facilitating infractions of academic integrity
- Plagiarism
- Bribes, favors, and threats
- Academic interference

- Examination by proxy
- Grade tampering
- Non-original works

Should an incident arise in which a student is thought to have violated academic integrity, the student will be processed under the disciplinary policy set forth in the Illinois Academic Integrity Policy. If you do not understand relevant definitions of academic infractions, contact your instructor for an explanation within the first week of class.

Collaboration, Copying, and Plagiarism

You are encouraged to discuss course material, including assignments, with your classmates. Homework assignments are meant to be learning experiences. You may discuss the exercises with other students, but you must write-up the solutions on your own. **In short, do not cheat, it is not worth the risk. You are more likely to get caught than you believe.** If you think you may be operating in a grey area, you most likely are. To ensure that you do not violate the academic integrity policies, all code and solutions should be prepared yourself. All code should be typed yourself, not copy and pasted. (Except for code provided in class.)

All work you turn in must be your own. This includes both written explanations, and code. Acknowledge any help received on your assignments. Copied assignments will receive no credit.

Copying from other students, books, websites, or solutions from previous versions of the class,

- does nothing to help you learn how to program,
- is easy for us to detect, and
- has serious negative consequences for you, as outlined in the university's policy on cheating and plagiarism which can be found in Article 1, Part 4 of the Student Code. Section 1-402 in particular outlines behavior which is considered an infraction of academic integrity.. If, after reading the policy, you are unclear on what is acceptable, please ask the instructor.

The homeworks and quizzes obviously must be done completely independently. Copying or any other form of cheating will result in a 0 for the exam, and possibly a more severe sanction depending on the circumstances.

Likewise, posting any course material on public sites like coursehero etc or making any course material available to other students is considered cheating and plagiarism. Please do not do it. Serious action will be taken.

Emergencies

If you have an emergency that will keep you from participating in the course, please notify your instructor by using the instructor's email address (listed on the Instructor Information page). Provide callback information in your email (if necessary). You should also notify your program director of any emergencies.

Sexual Misconduct Policy and Reporting

The University of Illinois is committed to combating sexual misconduct. Faculty and staff members are required to report any instances of sexual misconduct to the university's Title IX and Disability Office. In turn, an individual with the Title IX and Disability Office will provide information about rights and options, including accommodations, support services, the campus disciplinary process, and law enforcement options.

A list of the designated university employees who, as counselors, confidential advisors, and medical professionals, do not have this reporting responsibility and can maintain confidentiality, can be found in the Confidential Resources section. Other information about resources and reporting is available at wecare.illinois.edu.

Changes

The instructor reserves the right to make any changes he considers academically advisable. Such changes, if any, will be announced in class. Please note that it is your responsibility to follow the class and keep track of the proceedings.

COVID 19

Following University policy, all students are required to engage in appropriate behavior to protect the health and safety of the community. Students are also required to follow the campus COVID-19 protocols.

Students who feel ill must not come to class. In addition, students who test positive for COVID-19 or have had an exposure that requires testing and/or quarantine must not attend class. The University will provide information to the instructor, in a manner that complies with privacy laws, about students in these latter categories. These students are judged to have excused absences for the class period and should contact the instructor via email about making up the work.

Students who fail to abide by these rules will first be asked to comply; if they refuse, they will be required to leave the classroom immediately. If a student is asked to leave the classroom, the non-compliant student will be judged to have an unexcused absence and reported to the Office for Student Conflict Resolution for disciplinary action. Accumulation of non-compliance complaints against a student may result in dismissal from the University.

All students, faculty, staff, and visitors are required to wear face coverings in classrooms and university spaces. This is in accordance with CDC guidance and University policy and expected in this class.

Please refer to the University of Illinois Urbana-Champaign's COVID-19 website for further information on face coverings. Thank you for respecting all of our well-being so we can learn and interact together productively.

In order to implement COVID-19-related guidelines and policies affecting university operations, instructional faculty members may ask students in the classroom to show their Building Access Status in the Safer Illinois app or the Boarding Pass. Staff members may ask students in university offices to show their Building Access Status in the Safer Illinois app or the Boarding Pass. If the Building Access Status says "Granted," that means the individual is compliant with the

university's COVID-19 policies—either with a university-approved COVID-19 vaccine or with the on-campus COVID-19 testing program for unvaccinated students.

Students are required to show only the Building Access Screen, which shows compliance without specifying whether it was through COVID-19 vaccination or regular on-campus testing. To protect personal health information, this screen does not say if a person is vaccinated or not. Students are not required to show anyone the screen that displays their vaccination status. No university official, including faculty members, may ask students why they are not vaccinated or any other questions seeking personal health information.